



# Media reach and the market reaction to negative ESG media stories: do industry affiliation and ESG performance matter?<sup>☆</sup>

Siqi Liu<sup>a,1,\*</sup>, Wei Jiang<sup>b</sup>, Andrew W. Stark<sup>b</sup>

<sup>a</sup> The Open University Business School, The Open University, United Kingdom

<sup>b</sup> The University of Manchester Alliance Manchester Business School, United Kingdom

## ARTICLE INFO

### Keywords:

ESG negative media coverage  
Market reaction  
Media reach  
Industry affiliation  
ESG performance

## ABSTRACT

Using an event study of U.S.-listed firms, we examine how media reach conditions the stock market reactions to adverse environmental, social, and governance (ESG) news across firms with different ESG performance and industry affiliations. On average, we find that a one-unit increase in media reach is associated with a decrease of 0.13 percentage points in cumulative abnormal returns, indicating an economically meaningful valuation effect. Further analysis shows that this effect is mainly concentrated among low-ESG firms in controversial industries. For high-ESG firms in these industries, media reach makes stock price declines significantly less severe than for low-ESG firms, indicating a media reach insurance effect. However, unless adverse news receives the highest-reach media coverage, we do not find evidence that high ESG performance provides an overall insurance effect. By identifying media reach as a key moderator, this study refines the ESG insurance hypothesis and clarifies when markets value ESG performance.

## 1. Introduction

The integration of environmental, social, and governance (ESG) factors into corporate evaluations reflects increasing societal and investor expectations for sustainability and ethical conduct. ESG performance influences market perceptions, serving as both a signal of risk management and a response to societal pressures (Krüger, 2015; Capelle-Blancard & Petit, 2019). Negative ESG events, such as environmental violations or ethical breaches, can result in financial penalties and reputational damage, underscoring the potential importance of effective ESG strategies.

When adverse ESG news events occur, stock market participants must assess how the news affects their estimates of the levels and risks of future firm cash flows and, consequently, firm value. To do so, they need to fully understand the context in which the news takes place, including how stakeholders (such as customers, employees, suppliers) will react to the news and change their behavior, relative to the behavior implicitly expected in the current stock price. These behavioral changes will alter expected future cash flows, their risk, and the associated estimates of

market value. In turn, this will inform their trading behavior in response to the news. They might also have other reasons to trade (typically sell), as a consequence of nonfinancial concerns (for example, socially responsible funds concerned about ESG performance as a criterion for investment in a firm).

Prior studies have tested the market reaction to ESG-related news and, in general, their results are consistent: the stock market reacts negatively to adverse ESG-related news (Lundgren & Olsson, 2010; Becchetti et al., 2012; Flammer, 2013; Krüger, 2015; Capelle-Blancard & Petit, 2019; Boone & Uysal, 2020; De Vincentiis, 2024; Nicolas et al., 2024). Consistent with this general result, existing research finds that ESG reputational risk can, on average, have long-term consequences, reducing firm growth opportunities and market longevity (Fafaliou et al., 2022).

This raises a further question—if adverse ESG news produces, on average, a negative stock market reaction, does the reaction vary by type of firm (for example, high ESG performance firms versus low; firms in a controversial industry versus noncontroversial industry)? In this regard, some prior research documents that externally produced assessments of

<sup>☆</sup> This article is part of a special issue entitled: 'ESG Performance' published in Journal of Business Research.

\* Corresponding author.

E-mail address: [sl24668@open.ac.uk](mailto:sl24668@open.ac.uk) (S. Liu).

<sup>1</sup> Siqi Liu acknowledges the support of the China Scholarship Council while studying for her PhD at the Alliance Manchester School of the University of Manchester. Corresponding author: Dr. Siqi Liu, Walton Hall, Kents Hill, Milton Keynes, United Kingdom, MK7 6AA ([sl24668@open.ac.uk](mailto:sl24668@open.ac.uk)). The research time of Wei Jiang and Andrew Stark was paid for by general research funds from their respective institutions.

**Table 1**  
Studies testing the market reaction to ESG-related news.

| Authors                    | Year | Data                                                                                                                                 | Sample                                                                  | Key independent variable                                                                                                   |
|----------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Godfrey et al.             | 2009 | Wall Street Journal articles published between 1992 and 2003                                                                         | 185 events                                                              | CSR participation constructed from KLD data                                                                                |
| Lundgren and Olsson        | 2010 | 1260 firms monitored by GES in 2003–2006                                                                                             | The global firms covered by GES Investment service                      | Fama, French and Carhart four-factors                                                                                      |
| Becchetti et al.           | 2012 | Entries or exits from the Domini 400                                                                                                 | 327 events of entries or exits from the Domini 400 concerning 278 firms | Domini 400 Social Index                                                                                                    |
| Flammer                    | 2013 | Factiva, to search the Wall Street Journal (WSJ) for relevant press coverage                                                         | US firms covered by Factiva from 1980 to 2009                           | Environmental performance, as measured by KLD scores                                                                       |
| Krüger                     | 2015 | KLD socrates; @KLD newsletters                                                                                                       | US public firms covered by KLD socrates                                 | CSR performance: MSCI KLD data                                                                                             |
| Capelle-Blancard and Petit | 2019 | 33,000 ESG positive and negative news from Covalence EthicalQuote                                                                    | US firms from 2002 to 2010                                              | Firm reputation, economic crisis, media source etc.                                                                        |
| Boone and Uysal            | 2020 | KLD data                                                                                                                             | US listed firms, 20,610 firm-year observations from 1996 to 2011        | KLD environmental strengths and concerns                                                                                   |
| Wong and Zhang             | 2022 | Reprisk Rating                                                                                                                       | US publicly traded companies from 2007 to 2018                          | Change of Reprisk rating, firm size, S&P500 constituents, reputation status and FF48 (and gaming) industry classifications |
| Serafeim and Yoon          | 2022 | ESG news from Factset TruValue Lab                                                                                                   | US listed firms from 2010 to 2018                                       | ESG ratings: MSCI ESG Ratings, Sustainalytics and Thomson Reuters Asset 4                                                  |
| De Vincentiis              | 2024 | All the component firms of the euro Stoxx 50, Stoxx Europe 50, Stoxx Nordic 30, Stoxx USA 50 and Stoxx Asia/Pacific 50 stock indexes | Global firms's data from 2015 to 2020                                   | ESG performance comes from MSCI ESG Ratings                                                                                |
| Nicolas et al.             | 2024 | 114 million firm-related tweets for S&P 100 constituents                                                                             | S&P 100 constituents from 2016 to 2022                                  | Dummy variable for good news, dummy variable for bad news, critical ESG profile dummy                                      |
|                            |      |                                                                                                                                      |                                                                         | ESG controversy news from LSEG (Previously Refinitiv Eikon)                                                                |

high ESG performance can effectively function as a form of reputational insurance, or a “reservoir of goodwill,” that mitigates stock market reactions when adverse events occur (e.g., Godfrey et al., 2009; Cai et al., 2012; Lins et al., 2017; Shiu & Yang, 2017; Afrin et al., 2022; Pandey et al., 2024).

Nonetheless, the evidence on an insurance effect is not entirely unequivocal. For example, other studies (Krüger, 2015; Cui & Docherty, 2020; Wong & Zhang, 2022) find that the stock prices for firms with high ESG performance react more negatively to adverse ESG news than those for firms with low performance. Further, Eccles et al. (2022), in revisiting the performance of “sin stocks,” find that these firms do not differ from non-sin stocks in their announcement returns around negative ESG news.

Existing studies, however, rarely consider the media outlets from which adverse ESG news emanates, or how widely the news is disseminated by these media outlets (e.g., Cai et al., 2012; Capelle-Blancard & Petit, 2019; Krüger, 2015; Kölbel et al., 2017; Shiu & Yang, 2017; Afrin et al., 2022; Pandey et al., 2024). We ask—does this omission matter to the inferences that can be drawn about stock market reactions to adverse ESG news and the roles played by ESG performance and controversial industry affiliation?

We argue that all other things being equal, consistent with media agenda-setting theory (McCombs & Shaw, 1972), media outlets with greater reach increase information visibility, public scrutiny, and perceived reliability. Such increases in information visibility, public scrutiny, and perceived reliability could affect stakeholders, including investors, and their behavior with respect to the firm (e.g., more customers withdrawing or reducing their demand for a firm's products; more key personnel leaving the firm; and more fund managers selling their shareholdings in affected firms). Given our focus on negative ESG news, we argue that, on average, these effects could amplify any stock market reaction to such news. Further, in this context, an external assessment of high ESG performance—perhaps particularly for firms operating in controversial industries—could provide a degree of insurance against the amplification effect of media reach or, in contrast, could make the amplification effect even greater.

Building on these lines of argument, we examine how media reach affects the stock market reaction to negative ESG news, with a particular

focus on whether, and under what circumstances, high ESG performance mitigates the valuation losses associated with media reach following adverse ESG news. In particular, we investigate whether adverse ESG news reported by media outlets with different reach levels generate different stock price reactions for firms with high or low ESG performance and with or without controversial industry affiliation.

We use an event study methodology with data taken from the RepRisk database to examine the impact of negative ESG media coverage on stock returns. The RepRisk database provides precise dates for negative ESG news events, along with details such as media reach (indicating the influence of the media outlet) and severity level (assessing the potential cost or health and safety implications of the incident). To explore how these factors interact, we analyze subsamples based on the severity of the ESG news, the firm's level of information asymmetry (as captured by levels of share turnover), and whether the adverse ESG news refers to a breach of the UN Global Compact. By so doing, we investigate how these characteristics influence the relationship between media reach and stock market reactions.

This study provides novel evidence that the market value implications of ESG performance are generally affected by the reach of the media through which adverse ESG news is disseminated. We show that, on average, greater media reach amplifies stock price declines following adverse ESG events. However, this effect is heterogeneous across firms. In particular, our results suggest that the negative impact of media reach is mainly, although not exclusively, concentrated among firms in controversial industries with low ESG performance. In contrast, for firms in controversial industries with high ESG performance, the impact of media reach is generally insignificant and is significantly lower for these firms relative to their low ESG peers within controversial industry sectors. This suggests that high ESG performance can provide some insurance against the effect of media reach on market reactions to adverse ESG news. Taken together, these findings add to the general ESG performance “insurance effect” documented in prior research (e.g., Godfrey et al., 2009; Cai et al., 2012; Lins et al., 2017; Shiu & Yang, 2017; Afrin et al., 2022; Pandey et al., 2024). We also find, however, that high ESG performance only provides an overall insurance effect against adverse ESG news for firms in controversial industries if the information emanates from a high-reach media outlet.

These findings contribute to the ESG and financial markets literature in two ways. First, we introduce media reach as a previously overlooked but meaningful determinant of stock market reactions to adverse ESG news, thereby integrating media agenda-setting theory (McCombs & Shaw, 1972) into the study of ESG-related market reactions. Second, we refine the notion of high ESG performance as a reputational insurance mechanism by demonstrating that its protective effect is context-contingent rather than unconditional. Collectively, this study advances understanding of how ESG performance, media dynamics, and industry context jointly shape investor responses to negative ESG information.

This study has direct relevance for how firms, stock market participants, and ESG evaluators interpret the economic value of ESG performance when adverse ESG events occur. For managers, particularly in controversial industries, the results of the study imply that ESG engagement reduces downside risk only when adverse ESG news is reported by high-reach media outlets. For stock market participants, the results indicate that ESG scores should be interpreted in conjunction with the reach of the media outlets through which adverse ESG news is reported. Our results suggest that ESG performance provides meaningful overall downside protection for firms in controversial industries only when adverse ESG news is reported by high-reach media outlets. Evaluating ESG risk without accounting for media reach may overstate the protective value of ESG performance, particularly for firms operating in controversial sectors.

The rest of this paper is organized as follows. Section 2 presents the literature review and develops the hypotheses. Section 3 describes the methodology, and Section 4 details the data. Section 5 reports the main results, followed by robustness tests in Section 6. Finally, Section 7 summarizes the conclusions, discusses the study's limitations, and suggests directions for future research.

## 2. Previous literature, theoretical framework and hypotheses development

Much previous research on the market reaction to news, using event studies, focuses on general financial concerns (e.g., earnings announcements, stock splits) rather than news about ESG matters. As ESG has become a more important topic considered relevant to the value of firms (potentially through the broad channel of stakeholder behavior, as opposed to merely investors and shareholders), research has considered the market reaction to media coverage about ESG (or general CSR) topics using event studies (Godfrey et al., 2009; Lundgren & Olsson, 2010; Capelle-Blancard & Petit, 2019; Serafeim & Yoon, 2022). These studies test the market reaction to ESG-related news from different perspectives but, in general, their results are consistent. On average, the market reacts negatively to adverse ESG-related news (Lundgren & Olsson, 2010; Becchetti et al., 2012; Flammer, 2013; Krüger, 2015; Capelle-Blancard & Petit, 2019; Boone & Uysal, 2020; De Vincentiis, 2024; Nicolas et al., 2024). Table 1 summarizes this stream of the literature.

As shown in the table, previous studies that investigate the market reaction to ESG-related news do not pay much attention to the role of media reach. This may be due to a lack of data on media reach or the use of a single source of media coverage (for example, Flammer, 2013).<sup>2</sup>

As mentioned above, other prior research has interpreted high ESG performance as an “insurance effect” that protects firms from the stock market price effects of adverse events (Godfrey et al., 2009; Cai et al., 2012; Pandey et al., 2024). Nonetheless, the evidence on the effects of adverse ESG news on stock prices is not entirely unequivocal concerning the role played by the level of ESG performance. Expectations of ESG performance, and the likelihood of future adverse ESG news and their consequences already built into stock prices, play a role, as do the

patterns of stock ownership of affected firms.

For example, Cui and Docherty (2020) argue that high-ESG firms suffer more from bad news because, when a “good” company has a scandal, it violates investors’ trust, leading to a massive sell-off, with attendant stock price effects. Since low-ESG firms do not have this “trust” to begin with, they avoid this sharp sell-off. Wong and Zhang (2022) find that, while negative ESG media coverage significantly hurts stock prices in industries such as food, banking, and insurance, stocks in the “sin triumvirate” (alcohol, tobacco, and gaming) are not significantly affected by negative ESG news. This is because the stock price already reflects ESG reputational risk. Investors in a tobacco company, for example, are not surprised by news of negative health impacts, so they do not sell off the stock when such news breaks.

Some prior research also suggests that controversial industry affiliation shapes investor responses to adverse ESG news (e.g., Godfrey et al., 2009; Cai et al., 2012). Cai et al. (2012) define controversial industries as sectors characterized by persistent ethical or environmental concerns, including the alcohol, tobacco, gambling, oil, biotech, and cement sectors. Firms in these industries are more frequently exposed to negative ESG incidents and heightened media and investor scrutiny relative to noncontroversial sectors.

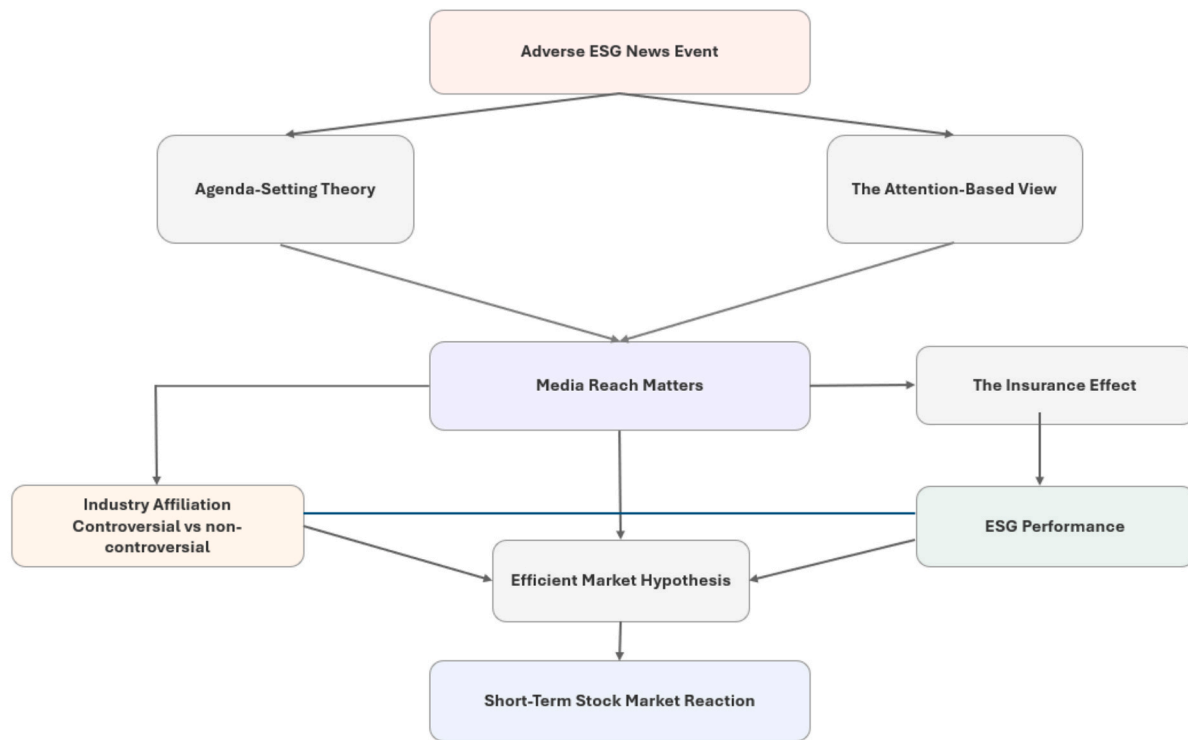
Further, firms in controversial industries may strategically invest in ESG activities to attempt to manage reputational risk and benefit from the insurance effect of ESG performance (Godfrey et al., 2009; Cai et al., 2012; Lins et al., 2017; Shiu & Yang, 2017; Afrin et al., 2022; Pandey et al., 2024). In doing so, sustained high ESG performance can represent a costly and credible signal of superior risk management, long-term orientation, and genuine commitment to stakeholders. For firms operating in controversial industries, maintaining high ESG performance may be particularly informative, as it requires greater effort and resources relative to firms in less scrutinized sectors.

Krüger (2015), however, argues that, because “sin stocks” (tobacco, gambling) have a low baseline of social expectations, a scandal is viewed as “business as usual,” resulting in a muted price effect compared to a “green” firm experiencing the same event. Eccles et al. (2022), in revisiting the performance of “sin stocks,” find that these firms do not differ from non-sin stocks in their announcement returns around negative ESG news. Their explanation is that “socially conscious” investors have already screened these stocks out of their portfolios, and the remaining shareholders (often hedge funds or income-focused investors) are indifferent to bad ESG headlines, leading to a muted price reaction.

Recent studies identify other circumstances under which ESG performance provides reputational insurance, and also factors that affect measures of ESG performance. Pandey et al. (2024) show that the adoption of mandatory ESG assurance in India generated negative market reactions, but firms with stronger ESG performance experienced significantly attenuated losses, consistent with a reservoir-of-goodwill mechanism. He et al. (2024) find that media coverage acts as an effective form of external, noninstitutional oversight that significantly improves corporate ESG performance, primarily by attracting greater analyst attention and reducing agency costs. Ali and Ali (2025) further suggest that negative media coverage pressure is associated with higher ESG rating divergence, based on evidence from Chinese firms. Research on the financial consequences of ESG engagement also highlights that the relationship between ESG initiatives and performance outcomes is far from uniform and is moderated by firm- and country-level factors. Recent studies further suggest that the financial implications of ESG performance are context-dependent. For example, Liaquat et al. (2026) find that ESG performance enhances bank stability, but that this relationship varies across national cultural and institutional settings. Similarly, Alamoudi and Hamoudah (2023) examine Saudi-listed banks and show that the association between ESG initiatives and financial performance is moderated by bank size. These findings highlight the importance of examining the conditions under which ESG performance affects financial outcomes.

Building on this research, and drawing on media agenda-setting

<sup>2</sup> Suryani et al. (2025) provide evidence from Indonesia of negative market reactions to adverse ESG news.



**Fig. 1.** Theoretical framework of the study. Note. The figure illustrates the proposed relationships among adverse ESG news events, media reach, ESG performance, industry affiliation, and stock market reactions. The framework draws on the attention-based view (Ocasio, 1997), agenda-setting theory (McCombs & Shaw, 1972), efficient market theory (Fama et al., 1969), and the ESG insurance hypothesis (Godfrey et al., 2009). ESG performance and industry affiliation are presented as contextual factors that shape investors' interpretation of adverse ESG news and moderate the effect of media reach on market reactions.

theory (McCombs & Shaw, 1972) and the attention-based view (Ocasio, 1997), we argue that the degree of media reach plays an important role in determining not only the breadth of information diffusion but also the intensity of cognitive attention that stakeholders (including stock market participants) allocate to a firm's ESG conduct. This affects their views of the firm and how they interact with it. These effects will be set in the context of what is known about ESG performance and affected firms' industry affiliations. On average, stakeholders are more likely to react to the news, whether favorably or unfavorably toward the firm, when it is disseminated by a media outlet with greater reach.

Fig. 1 below presents the theoretical framework of the study. It illustrates, through the lenses of agenda-setting theory and the attention-based view, the role of media reach in shaping short-term stock market reactions to adverse ESG news events. The figure also shows how industry affiliation and ESG performance are incorporated as key contextual factors that influence investors' interpretation of adverse ESG news.

Building on this theoretical framework, we develop testable hypotheses. Because media outlets differ in their ability to disseminate information, media reach likely shapes stakeholder responses to adverse ESG news. From an information-economics perspective, increased media reach reduces the cost of information searches and accelerates information diffusion, making stock market participants more aware of ESG incidents (Grossman & Stiglitz, 1980; Fang & Peress, 2009). More influential media will also likely affect the views of more stakeholders, feeding into stakeholder behavior with respect to their interactions with affected firms. Thus, all other things being equal, given the generally negative stock price reaction to negative ESG news events, adverse ESG news reported by high-reach media should produce more negative valuation effects relative to those reported by low-reach media – an amplification effect.

*H<sub>1</sub>. All else equal, the stock market reaction to adverse ESG news is more negative when the news is covered by higher-reach media.*

Next, we examine whether the effect of media reach on stock market reactions to negative ESG news varies across firms with different industry affiliations and ESG performance levels. As noted above, prior research suggests that both controversial industry affiliation and ESG performance shape investor responses to adverse ESG events. Firms in controversial industries are more frequently exposed to negative ESG incidents and heightened media and investor scrutiny relative to noncontroversial sectors. They may therefore strategically invest in ESG activities to manage reputational risk and mitigate the negative consequences of adverse events. This can work because sustained high ESG performance represents a costly and credible signal of superior risk management, long-term orientation, and genuine commitment to stakeholders. For firms operating in controversial industries, maintaining high ESG performance may be particularly informative, as it requires greater effort and resources relative to firms in less scrutinized sectors.

Nonetheless, the potential stock price reaction to negative ESG news is complex because existing perceptions of the firm, its ESG performance, and its industry status are already reflected in stock prices. When negative ESG news is covered by high-reach media, the same media attention that amplifies the impact of the adverse event may also enhance the visibility and credibility of a firm's established ESG record. Nonetheless, this could attenuate or increase the negative stock market response. Given the range of possibilities, we state the second hypothesis in three different forms:

*H<sub>2</sub>: All other things being equal, the relationship between media reach and the stock market reaction to adverse ESG news is different for high ESG performance firms in controversial industries relative to firms in controversial industries with low ESG performance.*

*H<sub>2a</sub>: All other things being equal, the relationship between media reach and the stock market reaction to adverse ESG news is more negative for high ESG performance firms in controversial industries relative to firms in controversial industries with low ESG performance.*



**Table 2**  
Total Media reach effects for different sector/ESG performance combinations.

|                            | Low ESG performance   | High ESG performance                      | Difference            |
|----------------------------|-----------------------|-------------------------------------------|-----------------------|
| Non-controversial Industry | $\beta_1$             | $(\beta_1 + \beta_3)$                     | $\beta_3$             |
| Controversial industry     | $(\beta_1 + \beta_2)$ | $(\beta_1 + \beta_2 + \beta_3 + \beta_5)$ | $(\beta_3 + \beta_5)$ |
| Difference                 | $\beta_2$             | $(\beta_2 + \beta_5)$                     |                       |

**Table 3**  
Variable descriptions and sources.

| Dependent variable     |                                                                                                                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variables              | Variable description                                                                                                                                                                                                                                                   |
| CAR                    | The cumulative abnormal return over the days $(-1,1)$ , where day 0 is the announcement date, and using the equally-weighted CRSP index as the benchmark return                                                                                                        |
| Experimental variables |                                                                                                                                                                                                                                                                        |
| Reach                  | A categorical variable, equal to 1,2 or 3, which denotes whether the source for the media event is of low, medium or high reach level, taken from Reprisk                                                                                                              |
| Controversial Industry | A dummy variable, equal to 1 if a firm belongs to a controversial industry (including tobacco, alcohol, gambling, oil, biotech, cement industries); otherwise equals 0                                                                                                 |
| ESG                    | Corporate social responsibility performance in the year prior to the news media event, using yearly data from Thomson Reuters                                                                                                                                          |
| High ESG               | A dummy variable, equal to 1 if the firm has a higher ESG performance score than the median of ESG scores in year t-1, otherwise 0                                                                                                                                     |
| Control variables      |                                                                                                                                                                                                                                                                        |
| Severity               | A categorical variable, equal to 1, 2 or 3, which denotes whether the ESG controversy media event is of low, medium or high severity. If the severity is 2 or 3, the ESG negative event is of high severity level, otherwise it is of low severity level, from Reprisk |
| Novelty                | A categorical variable, equal to 2 if the media event describes an ESG incident for the first time, otherwise equals 1, from Reprisk                                                                                                                                   |
| Breach                 | A dummy variable, equal to 1 if the media event describes an ESG incident that breaches the United Nations Global Compact principles, otherwise equals 0, from Reprisk                                                                                                 |
| Peak RRI               | The log of market value at the highest level of the Reprisk Index over the last two years. It is a measurement of the reputational risk for a firm, from Reprisk                                                                                                       |
| Size                   | The log of market value at the end of the calendar year prior to the media event, from Compustat                                                                                                                                                                       |
| Lev                    | Leverage – the ratio of total liabilities to total assets at the end of the financial year prior to the media event, from Compustat                                                                                                                                    |
| ROA                    | Return on assets – the ratio of net income to total assets at the end of the financial year prior to the media event, from Compustat                                                                                                                                   |
| Sales Growth           | Sales growth – the ratio of the change in sales revenue in year t to sales revenue in year t-1, from Compustat                                                                                                                                                         |
| PPE                    | The ratio of property, plant and equipment to total assets at the end of the financial year prior to the media event, from Compustat                                                                                                                                   |
| Turnover               | The ratio of share trading volume to outstanding shares in the calendar year prior to the media event, from CRSP                                                                                                                                                       |
| High Turnover          | A dummy variable, equal to 1 if share turnover is higher than the median of turnover for the sample in year t-1, otherwise equals 0                                                                                                                                    |

H<sub>2b</sub>: All other things being equal, the relationship between media reach and the stock market reaction to adverse ESG news is less negative for high ESG performance firms in controversial industries relative to firms in controversial industries with low ESG performance.

Our hypotheses do not provide specific predictions about media reach effects for individual combinations of ESG performance and controversial/noncontroversial industry affiliation. Rather, they concentrate on either general media reach effects or relative media reach effects between high and low ESG performance for controversial

industry firms. We raise a possibility about such effects, however. When a firm in a controversial industry with high ESG performance faces negative ESG news, the media attention may serve a dual function. While it amplifies the news itself, it could also amplify the visibility of the firm's established ESG credentials. Under these conditions, the market may interpret the firm's high ESG performance not merely as a risk-mitigating 'insurance' asset but as a signal of superior managerial foresight, genuine stakeholder commitment, and enhanced resilience. This seems more likely the higher the level of media reach. This could attenuate any negative media reach effect so significantly that it results in a net neutral or even a positive market response, as investors reassess and potentially reward the firm's differentiated standing amid controversy. This could be interpreted as a positive 'reputational dividend' or affirmative signal.

Overall, our study extends prior studies in two important ways. First, it offers a potential reconceptualization of the "insurance effect" associated with high ESG performance, if it exists, as not a universal protective shield but as a conditional effect—one that depends on the level of media reach and the context of the industry. Second, focusing on industry context in the context of studying the impact of media reach allows for the possibility that high ESG performance is not always equally important in evaluating changes in expected future firm cash flows and the risks associated with those cash flows, and the likelihood of stock sell-offs stemming from adverse ESG news. In sum, this study establishes media reach as an important variable in understanding the impact of adverse ESG news. By specifying conditions for any insurance effect of high ESG performance to hold, it extends beyond the replication of prior findings to offering a more nuanced interpretation.

### 3. Research design

As mentioned above, our research employs an event study methodology to examine how negative ESG-related media coverage affects stock prices and returns around the news event. Theoretically, under the assumption of semi-strong market efficiency, stock market prices will react quickly to new information and in an unbiased fashion (Fama et al., 1969). Ideally, event studies work best when a shorter window around the event date is employed to capture the stock market reaction (Kothari & Warner, 2007) to avoid noise entering into the measure. Using a small event window in such analyses requires the precise date of such incidents or events to be available. The topic of negative ESG media coverage is suitable for the use of a short event window because the RepRisk database from which we derive our negative ESG news events identifies the date of the adverse media coverage.

To capture the immediate market reaction, we focus on a  $(-1, 1)$  event window, where day 0 corresponds to the publication date of the media story. Prior ESG event-study research shows that reactions to negative ESG or CSR news are concentrated around the announcement day, with little delayed adjustment (Krüger, 2015; Capelle-Blancard & Petit, 2019). A wider window introduces unrelated return noise, reducing identification precision. Therefore, a  $(-1, 1)$  event window is a theoretically and empirically justified window for capturing the market response to ESG news.

In an event study, the dependent variable is generally the cumulative abnormal return within a specific event window. Based on Fama et al. (1969), the calculation of the cumulative abnormal return (CAR) is as follows. First, the parameters of the market model,  $\alpha$  and  $\beta$ , are estimated in a pre-event period. The market model is described below:

$$R_{jt} = \alpha_j + \beta_j R_{mt} + \epsilon_{jt} \quad (1)$$

where  $R_{jt}$  is the return of common stock of the  $j^{\text{th}}$  firm on day t;  $R_{mt}$  is the rate of return of the market index on day t; and  $\epsilon_{jt}$  is a random error term.

The abnormal return,  $A_{jt}$ , for the common stock of  $j^{\text{th}}$  firm on day t is calculated as follows.

$$A_{jt} = R_{jt} - (\hat{\alpha}_j + \hat{\beta}_j R_{mt}) \quad (2)$$

where  $\hat{\alpha}_j$  and  $\hat{\beta}_j$  are ordinary least squares estimates of  $\alpha_j$  and  $\beta_j$ . The cumulative abnormal return  $CAR_{T_1, T_2}$  over the period  $(T_1, T_2)$  for  $N$  events is as follows:

$$CAR_{T_1, T_2} = \sum_{t=T_1}^{T_2} A_{j,t} \quad (3)$$

We compute cumulative abnormal returns using the Eventus package. Our main dependent variable, CAR, is the cumulative abnormal return over the  $(-1, 1)$  window, calculated using the equally weighted CRSP index as the market benchmark. For robustness, we also construct alternative measures using the value-weighted CRSP index in the market model over the same window (CAR1), the Fama–French–Carhart four-factor model (CAR2), and a wider  $(-2, 2)$  event window using equally weighted (CAR3) and value-weighted (CAR4) CRSP indices to allow for information leakage and delayed price adjustment.

We use Eq. (4) below as our benchmark model to test the average impact of media reach (*Reach*) on the cumulative abnormal return for all media events in the sample.

$$\begin{aligned} CAR = & \alpha_1 Reach + \alpha_2 Breach + \alpha_3 Severity + \alpha_4 Novelty + \alpha_5 PeakRRI_{t-1} \\ & + \alpha_6 ESG_{t-1} + \alpha_7 ControversialIndustry + \alpha_8 Size_{t-1} + \alpha_9 ROA_{t-1} \\ & + \alpha_{10} Lev_{t-1} + \alpha_{11} PPE_{t-1} + \alpha_{12} SalesGrowth_{t-1} + \sum_{j=1}^{46} \mu_j Ind_j \end{aligned} \quad (4)$$

The coefficient of interest for Hypothesis 1 is  $\alpha_2$ , the coefficient of *Reach*. Hypothesis 1 predicts that  $\alpha_1 < 0$ .

We then modify Eq. (4) to allow for the interactions implied by our second hypothesis.

$$\begin{aligned} CAR = & \beta_1 Reach + \beta_2 Reach \cdot ControversialIndustry + \beta_3 Reach \cdot HighESG \\ & + \beta_4 ControversialIndustry \cdot HighESG \\ & + \beta_5 Reach \cdot ControversialIndustry \cdot HighESG + \beta_6 ControversialIndustry \\ & + \beta_7 HighESG + \beta_8 Breach + \beta_9 Severity + \beta_{10} Novelty + \beta_{11} PeakRRI_{t-1} \\ & + \beta_{12} Size_{t-1} + \beta_{13} ROA_{t-1} + \beta_{14} Lev_{t-1} + \beta_{15} PPE_{t-1} \\ & + \beta_{16} SalesGrowth_{t-1} + \sum_{j=1}^{46} \gamma_j Ind_j \end{aligned} \quad (5)$$

We can use Eq. (5), and various coefficients and combinations of coefficients, to test our second hypothesis.

Table 2 details: (i) how we combine coefficients estimated through Eq. (5) to produce total incremental media reach effects for each combination of the dimensions of controversial/noncontroversial industry and high/low ESG performance; and (ii) how we test for differences along the dimensions. Table 3 details how we define the variables included in our regressions.

We broadly follow Kölbel et al. (2017) and Burke et al. (2019) with respect to the choice of control variables. Firm size (*Size*), return on assets (*ROA*), property, plant and equipment (*PPE*), and sales growth (*Sales Growth*) are included to capture different dimensions of firm visibility and operational characteristics that may influence media attention and investor response, following established practice (e.g., Bansal & Clelland, 2004). Leverage (*Lev*) controls for capital structure, which may affect financial risk, according to Kölbel et al. (2017). All regressions include Fama–French 48 industry fixed effects to control for unobserved industry-level heterogeneity.

We also include variables capturing: (i) the severity of the news story (*Severity*); (ii) the novelty of the news story (*Novelty*); (iii) whether the event involves a breach of the UN Global Compact (*Breach*); and (iv) a measure of firm reputation (*Peak RRI*). These control for other factors concerning the characteristics of the news event and the affected firm

that might affect the market reaction to the story.

We also conduct subsample analyses based on Eq. (5). First, we examine whether the results differ between low- and high-severity ESG news. Second, we partition firms by stock market liquidity, measured by share turnover, to account for differences in how efficiently firms incorporate economically relevant information. Finally, we conduct subsample analyses based on whether the negative ESG news involves breaches of the UN Global Compact (UNGC). We do so because we expect the effects of media reach to be more pronounced for high-severity or UN Global Compact breach stories and, in general, for high stock turnover (stock-market liquidity) stocks.

The impact of media reach could depend on the severity of ESG incidents or whether they constitute breaches of the UN Global Compact, because these incident features are likely to alter how stakeholders process media-borne information by affecting its perceived importance and emotional salience. This, in turn, will feed through into changes in their behavior toward the firm as a result of the incident and changes in investors' expectations of the firm's future cash flows and their risk. Low-severity or non-breach incidents could also be interpreted as isolated or manageable issues, whereas high-severity incidents heighten perceived reputational and financial risks, prompting stakeholders to treat the latter as credible and urgent. Consequently, severity or breach could function as amplifiers that condition the effect of media reach in shaping market reactions. With respect to creating subsamples based on share turnover, more highly traded shares tend to reflect price-relevant information more quickly than those that are less highly traded.

#### 4. Data

Our initial sample consists of U.S. publicly traded firms covered by the RepRisk database from 2007 to 2018. RepRisk is a Zurich-based provider specializing in ESG and responsible business conduct risks. Its news database identifies firm-level negative ESG incidents and records key attributes of each event, including its media reach and severity and whether it refers to a breach of the UN Global Compact. We obtain ESG news data from RepRisk at the firm–incident level, where each observation corresponds to a single RepRisk story ID associated with a firm on a given day. RepRisk consolidates duplicate coverage across multiple media sources into one incident and retains the most relevant content from the most influential source. When the same story appears in multiple languages, the English version is typically used.

From RepRisk, we also capture the following adverse news event characteristics. First, media reach (*Reach*) captures the influence of the reporting source and is coded as low (1), medium (2), or high (3), with high reach corresponding to influential international outlets such as the BBC or The New York Times. Second, severity (*Severity*) reflects the seriousness of the ESG incident based on its consequences, scope of impact, and underlying cause (e.g., accident, negligence, or intent), and is similarly classified into low, medium, or high levels. For illustration, tax optimization involving Starbucks is classified as low severity, financial fraud at Cisco Systems as medium severity, and human rights violations linked to Caterpillar as high severity. RepRisk also flags whether an incident constitutes a breach of the United Nations Global Compact.

In addition, it provides the Peak RepRisk Index (*Peak RRI*) as a summary measure of a firm's recent ESG-related reputational risk. *Peak RRI* is defined as the maximum RepRisk Index value over the preceding two years and captures cumulative reputational exposure to negative ESG and business conduct events. The index ranges from 0 to 100, with higher values indicating greater exposure to negative ESG-related media coverage. The RepRisk Index itself reflects a firm's current level of ESG-related media and stakeholder attention and is constructed based on the reach, severity, and novelty of news events, scaled by the intensity of ESG-related coverage in the preceding two months.

To avoid confounding effects, we exclude observations with overlapping ESG events for the same firm within a two-day window, as well

**Table 4**  
Sample derivation.

| Data merging steps                                  | Observations |
|-----------------------------------------------------|--------------|
| Reprisk news data (firm-incident level)             | 40,788       |
| After dropping observations with overlapping events | 20,968       |
| After dropping contaminated events                  | 17,400       |
| With stock market data available from CRSP          | 17,374       |
| With Compustat data available                       | 17,373       |
| With ESG score data                                 | 5077         |

**Notes:**

- (i) The Reprisk news data (firm-incident level) counts the number of observations when a firm has only one news event corresponding to one Reprisk story ID on a single day;
- (ii) overlapping events that are dropped denote those events for the same firm within 2 days and all observations with multiple events for a single firm on a single day; and.
- (iii) the contaminated events that are dropped include earnings announcements (1469), M&A announcements (694), SEO announcements (631), CEO turnover and directors (Chairman, President, CFO, CEO) appointments (758), and share repurchase (15).

as cases where multiple events occur for a firm on the same day. We further remove observations contaminated by other major corporate events during the event window, including earnings announcements, mergers and acquisitions, seasoned equity offerings, executive and

director appointments, and share repurchases (the importance of doing so is stressed in [De Jong & Naumovska, 2016](#)).

We then merge the remaining ESG incident data with cumulative abnormal returns from CRSP and Eventus and with firm-level control variables from Compustat measured in year  $t-1$ . This procedure yields 15,036 firm-incident observations. This sample is then merged with ESG performance scores from Thomson Reuters. Their ESG database dates back to fiscal year 2002, covering more than 6000 firms globally, but with most of the firms located in the U.S. and Europe. The ESG score measures a company's performance on environmental, social, and governance issues, based on reported data in the public domain. Specifically, the data are obtained from six sources: annual reports, company websites, NGO websites, stock exchange filings, CSR reports, and news sources. All indicators are grouped into three categories: environmental, social, and governance issues. The ESG score represents the firm's relative ESG performance. The higher the ESG score, the better the ESG performance.

The final sample comprises 5077 observations. Out of the Fama–French 48 industries, 46 are represented in our sample, as no firms belong to the soda or fabricated products industries. All regressions include Fama–French 48 industry fixed effects to control for unobserved industry-level heterogeneity. Detailed sample construction procedures are reported in [Table 4](#).

**Table 5**  
Descriptive statistics.

| Variables              | Mean   | Std. Dev. | Skewness | Kurtosis | Min    | P25    | P50    | P75    | Max    |
|------------------------|--------|-----------|----------|----------|--------|--------|--------|--------|--------|
| CAR                    | 0.000  | 0.032     | 0.608    | 55.674   | −0.461 | −0.013 | 0.000  | 0.012  | 0.544  |
| Reach                  | 1.812  | 0.759     | 0.329    | 1.797    | 1      | 1      | 2      | 2      | 3      |
| Controversial Industry | 0.207  | 0.405     | 1.446    | 3.092    | 0      | 0      | 0      | 0      | 1      |
| ESG                    | 55.872 | 16.223    | −0.140   | 2.195    | 10.340 | 43.610 | 55.960 | 68.580 | 93.170 |
| Severity               | 1.449  | 0.574     | 0.851    | 2.720    | 1      | 1      | 1      | 2      | 3      |
| Novelty                | 1.734  | 0.442     | −1.057   | 2.118    | 1      | 1      | 2      | 2      | 2      |
| Breach                 | 0.607  | 0.488     | −0.439   | 1.193    | 0      | 0      | 1      | 1      | 1      |
| Peak RRI               | 41.650 | 14.104    | 0.106    | 2.794    | 0      | 32     | 40     | 52     | 79     |
| Size                   | 9.856  | 1.421     | −0.303   | 2.844    | 5.805  | 8.990  | 9.851  | 10.904 | 12.833 |
| Lev                    | 0.654  | 0.196     | 0.520    | 3.944    | 0.192  | 0.515  | 0.643  | 0.768  | 1.318  |
| ROA                    | 0.056  | 0.073     | −0.595   | 5.748    | −0.215 | 0.022  | 0.054  | 0.091  | 0.237  |
| Sales Growth           | 0.039  | 0.161     | 0.656    | 6.955    | −0.452 | −0.032 | 0.033  | 0.097  | 0.779  |
| PPE                    | 0.323  | 0.254     | 0.629    | 2.264    | 0.004  | 0.109  | 0.250  | 0.525  | 0.890  |
| Turnover               | 0.209  | 0.153     | 2.618    | 12.135   | 0.041  | 0.113  | 0.167  | 0.259  | 0.982  |

**Notes:** All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. Skewness and kurtosis statistics are reported to help assess distributional properties.

**Table 6**  
Correlations.

| Variables              | CAR     | Reach    | Controversial Industry | ESG     | Severity | Novelty      |
|------------------------|---------|----------|------------------------|---------|----------|--------------|
| Reach                  | −0.041* |          |                        |         |          |              |
| Controversial Industry | 0.009   | 0.035*   |                        |         |          |              |
| ESG                    | 0.015   | 0.033*   | −0.045*                |         |          |              |
| Severity               | −0.005  | 0.007    | −0.018                 | −0.015  |          |              |
| Novelty                | 0.027   | −0.152*  | −0.055*                | 0.034*  | 0.179*   |              |
| Breach                 | −0.007  | −0.005   | 0.033*                 | −0.006  | 0.124*   | 0.101*       |
| Peak RRI               | −0.014  | 0.217*   | 0.270*                 | −0.028* | 0.072*   | −0.194*      |
| Size                   | −0.007  | 0.134*   | 0.287*                 | 0.082*  | 0.021    | −0.083*      |
| Lev                    | 0.008   | −0.016   | −0.091*                | 0.013   | −0.009   | 0.029*       |
| ROA                    | −0.036* | 0.081*   | 0.123*                 | 0.004   | 0.028*   | −0.024       |
| Sales Growth           | −0.027  | 0.037*   | 0.031*                 | −0.096* | −0.058*  | −0.014       |
| PPE                    | 0.027   | −0.106*  | 0.166*                 | −0.013  | −0.016   | 0.005        |
| Variables              | Breach  | Peak RRI | Size                   | Lev     | ROA      | Sales Growth |
| Peak RRI               | 0.045*  |          |                        |         |          |              |
| Size                   | −0.080* | 0.559*   |                        |         |          |              |
| Lev                    | −0.078* | −0.023   | −0.022                 |         |          |              |
| ROA                    | −0.088* | 0.204*   | 0.446*                 | −0.199* |          |              |
| Sales Growth           | −0.012  | −0.004   | 0.046*                 | −0.170* | 0.191*   |              |
| PPE                    | 0.299*  | −0.081*  | −0.202*                | −0.037* | −0.236*  | 0.027        |

\* Shows significance at the 0.05 level.

**Table 7**

Market reactions to negative ESG media events.

| Variables                             | Dependent Variable – CAR |                      |                      |                     |                     |                      |                      |                     |
|---------------------------------------|--------------------------|----------------------|----------------------|---------------------|---------------------|----------------------|----------------------|---------------------|
|                                       | Baseline model           | Full sample          | Low severity         | High Severity       | Low Turnover        | High Turnover        | Breach               | Non-Breach          |
| Reach                                 | −0.130**<br>(0.048)      | −0.091<br>(0.295)    | 0.009<br>(0.941)     | −0.237**<br>(0.031) | 0.070<br>(0.580)    | −0.236**<br>(0.012)  | −0.117<br>(0.291)    | −0.044<br>(0.723)   |
| Controversial Industry                | 0.396<br>(0.431)         | 1.222*<br>(0.058)    | 3.713***<br>(0.000)  | −0.569<br>(0.486)   | 1.543*<br>(0.057)   | 1.139<br>(0.104)     | 0.188<br>(0.825)     | 2.465***<br>(0.003) |
| ESG Score                             | 0.002<br>(0.477)         |                      |                      |                     |                     |                      |                      |                     |
| High ESG                              |                          | 0.005<br>(0.983)     | 0.316<br>(0.267)     | −0.360<br>(0.247)   | −0.094<br>(0.720)   | 0.140<br>(0.644)     | 0.182<br>(0.494)     | −0.242<br>(0.543)   |
| Controversial Industry.High ESG       |                          | −1.159***<br>(0.010) | −1.528***<br>(0.005) | −0.779*<br>(0.086)  | −0.730**<br>(0.016) | −1.739***<br>(0.000) | −1.926***<br>(0.000) | 0.472<br>(0.531)    |
| Reach.Controversial Industry          |                          | −0.460**<br>(0.033)  | −0.651**<br>(0.012)  | −0.243<br>(0.271)   | −0.399**<br>(0.027) | −0.591***<br>(0.000) | −0.573***<br>(0.003) | −0.169<br>(0.314)   |
| Reach.High ESG                        |                          | −0.014<br>(0.906)    | −0.159<br>(0.299)    | 0.191<br>(0.193)    | 0.036<br>(0.785)    | −0.070<br>(0.667)    | −0.05<br>(0.694)     | 0.055<br>(0.784)    |
| Reach.Controversial Industry.High ESG |                          | 0.703***<br>(0.002)  | 0.893***<br>(0.002)  | 0.468**<br>(0.021)  | 0.517**<br>(0.026)  | 0.898***<br>(0.000)  | 1.000***<br>(0.000)  | 0.108<br>(0.770)    |
| Breach                                | −0.100<br>(0.305)        | −0.098<br>(0.309)    | −0.058<br>(0.576)    | −0.127<br>(0.416)   | 0.003<br>(0.980)    | −0.216<br>(0.146)    |                      |                     |
| Severity                              | −0.030<br>(0.599)        | −0.020<br>(0.726)    |                      |                     | 0.035<br>(0.653)    | −0.026<br>(0.801)    | 0.013<br>(0.850)     | −0.043<br>(0.661)   |
| Novelty                               | 0.104<br>(0.193)         | 0.100<br>(0.216)     | 0.130<br>(0.143)     | 0.009<br>(0.947)    | −0.012<br>(0.850)   | 0.229<br>(0.123)     | 0.170*<br>(0.059)    | −0.013<br>(0.911)   |
| Peak RRI                              | 0.001<br>(0.829)         | 0.001<br>(0.778)     | 0.001<br>(0.768)     | 0.002<br>(0.809)    | 0.003<br>(0.320)    | 0.005<br>(0.438)     | 0.002<br>(0.657)     | 0.000<br>(0.999)    |
| Size                                  | 0.018<br>(0.729)         | 0.018<br>(0.723)     | −0.020<br>(0.721)    | 0.069<br>(0.353)    | −0.12<br>(0.126)    | 0.076<br>(0.276)     | 0.007<br>(0.934)     | 0.035<br>(0.670)    |
| ROA                                   | −1.515**<br>(0.046)      | −1.454**<br>(0.040)  | −1.142<br>(0.262)    | −1.554<br>(0.102)   | −0.278<br>(0.811)   | −1.685**<br>(0.039)  | −1.056<br>(0.245)    | −1.638<br>(0.126)   |
| Lev                                   | −0.225<br>(0.310)        | −0.251<br>(0.267)    | −0.217<br>(0.536)    | −0.218<br>(0.557)   | −0.169<br>(0.709)   | −0.110<br>(0.744)    | −0.416<br>(0.194)    | −0.093<br>(0.777)   |
| PPE                                   | 0.509**<br>(0.031)       | 0.438*<br>(0.050)    | 0.058<br>(0.841)     | 0.994**<br>(0.022)  | 0.043<br>(0.900)    | 0.773**<br>(0.016)   | 0.533<br>(0.104)     | 0.61<br>(0.315)     |
| Sales Growth                          | −0.374<br>(0.146)        | −0.424<br>(0.104)    | −0.009<br>(0.980)    | −1.169**<br>(0.015) | −0.299<br>(0.588)   | −0.420<br>(0.176)    | −0.651*<br>(0.059)   | −0.034<br>(0.937)   |
| Adj. R-square                         | 0.016                    | 0.019                | 0.024                | 0.038               | 0.020               | 0.039                | 0.028                | 0.041               |
| F statistics                          | 1.442                    | 1.579                | 1.207                | 1.335               | 0.919               | 1.65                 | 1.49                 | 1.44                |
| p value (F)                           | 0.018                    | 0.003                | 0.135                | 0.048               | 0.644               | 0.001                | 0.01                 | 0.018               |
| Observations                          | 5,077                    |                      | 3,006                | 2,071               | 2,542               | 2,535                | 3,083                | 1,994               |

Notes: This table presents the results for the market reaction to the negative ESG media coverage using Eqs. (4) and (5). Column 2 is for the results for the baseline Eq. (4), and the others show the results for Eq. (5). Among them, Column 3 presents the results for Eq. (5) using full sample. Columns 4 and 5 present the regression results for Eq. (5) using Low Severity and High Severity sub-samples. Columns 6 and 7 present the regression results for Eq. (5) using Low Turnover and High Turnover sub-samples. Columns 8 and 9 present the regression results for Eq. (5) using Breach and non-Breach sub-samples. p-values are in parentheses. All models are estimated without a constant term and include 45 industry dummies variables.

Descriptive statistics for the various variables are reported in Table 5. The table includes the minimum, maximum, mean, median, and standard error for each variable. Correlations between the variables are included in Table 6. The values for the correlations do not indicate potential multicollinearity problems for our regression coefficient estimates.<sup>3</sup>

## 5. Results

Table 7 reports the results for market reactions to negative ESG media coverage using Eqs. (4) and (5), while Table 8 presents F-test coefficients and p-values assessing the impact of media reach across firm types and testing differences between groups, as illustrated in Table 2. The low adjusted R-squared values are not unexpected in event-study settings. Short-window CARs are designed to capture market reactions

to specific news events, while a substantial proportion of cross-sectional return variation remains unexplained. As a result, cross-sectional CAR regressions often have limited explanatory power, and low adjusted R-squared values are common in this literature (Kothari & Warner, 2007; Moeller et al., 2004; De Jong & Naumovska, 2016). Accordingly, the validity of our conclusions rests primarily on the sign, magnitude, and statistical significance of the estimated effects rather than on the adjusted R-squared values of the estimated models. Nonetheless, F-tests reported in Table 7 confirm that the main models are jointly significant, other than for the low-severity and low-turnover subsample. In the subsequent discussion, we focus on the media reach effects for the four combinations of controversial/noncontroversial industry affiliation and high/low ESG performance.

The baseline results indicate that, on average, greater media reach is associated with more negative stock market reactions to adverse ESG news. Consistent with Hypothesis 1, this finding suggests that, on average, higher-reach media coverage amplifies the negative valuation impact of negative ESG events. This is consistent with the idea that, if negative ESG news is disseminated through highly influential media outlets, this increases information visibility, public scrutiny, and the perceived reliability of the information. These effects influence the behavior of more stakeholders and may adversely affect firm value.

Importantly, however, with respect to Hypothesis 2 in its various

<sup>3</sup> To assess potential multicollinearity concerns, we conducted variance inflation factor (VIF) diagnostics for all specifications reported in Tables 7 and 8. In particular, the VIF for Peak RRI is consistently around 2 across regressions, well below conventional thresholds of concern. We further re-estimated the VIFs in specifications excluding industry fixed effects but retaining a constant term and again find that the VIF for Peak RRI remains below 2 in all cases.



**Table 8**  
Total Coefficients of Media Reach.

| FULL SAMPLE                |                      |                     |                     |
|----------------------------|----------------------|---------------------|---------------------|
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.091<br>(0.295)    | −0.105<br>(0.215)   | −0.014<br>(0.906)   |
| Controversial Industry     | −0.551***<br>(0.006) | 0.138<br>(0.342)    | 0.689***<br>(0.000) |
| Difference                 | −0.460**<br>(0.033)  | 0.243<br>(0.159)    |                     |
| LOW SEVERITY SUB-SAMPLE    |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | 0.009<br>(0.941)     | −0.150<br>(0.177)   | −0.159<br>(0.299)   |
| Controversial Industry     | −0.645***<br>(0.005) | 0.092<br>(0.403)    | 0.734***<br>(0.003) |
| Difference                 | −0.651**<br>(0.012)  | 0.242<br>(0.126)    |                     |
| HIGH SEVERITY SUB-SAMPLE   |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.237**<br>(0.031)  | −0.046<br>(0.601)   | 0.191<br>(0.193)    |
| Controversial Industry     | −0.480**<br>(0.010)  | 0.179<br>(0.394)    | 0.659***<br>(0.000) |
| Difference                 | −0.243<br>(0.271)    | 0.225<br>(0.344)    |                     |
| LOW TURNOVER SUB-SAMPLE    |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | 0.070<br>(0.580)     | 0.106<br>(0.140)    | 0.036<br>(0.785)    |
| Controversial Industry     | −0.329**<br>(0.014)  | 0.224<br>(0.070)    | 0.553***<br>(0.006) |
| Difference                 | −0.399**<br>(0.027)  | 0.118<br>(0.406)    |                     |
| HIGH TURNOVER SUB-SAMPLE   |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.236**<br>(0.012)  | −0.306**<br>(0.047) | −0.070<br>(0.667)   |
| Controversial Industry     | −0.827***<br>(0.000) | 0.001<br>(0.993)    | 0.828***<br>(0.000) |
| Difference                 | −0.591***<br>(0.000) | 0.307<br>(0.097)    |                     |
| BREACH SUB-SAMPLE          |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.117<br>(0.291)    | −0.167<br>(0.128)   | −0.050<br>(0.694)   |
| Controversial Industry     | −0.690***<br>(0.000) | 0.260<br>(0.138)    | 0.950***<br>(0.000) |
| Difference                 | −0.573***<br>(0.003) | 0.427**<br>(0.039)  |                     |
| NON-BREACH SUB-SAMPLE      |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.044<br>(0.723)    | 0.011<br>(0.939)    | 0.055<br>(0.784)    |
| Controversial Industry     | −0.213<br>(0.109)    | −0.050<br>(0.846)   | 0.163<br>(0.615)    |
| Difference                 | −0.169<br>(0.314)    | −0.061<br>(0.839)   |                     |

Notes: This table presents the coefficients and p-values of F tests for each type of firms based on the results in Table 7. p-values are in parentheses.

forms, the impact of media reach is not uniform across firms but differs systematically by industry context and ESG performance. In particular, evidence from Table 8 shows that market reactions to adverse ESG news differ significantly between controversial industry firms with low versus high ESG performance. Specifically, the difference in the media reach effect between these two groups (the lower right difference estimates in

**Table 9**  
Market reactions to negative ESG media events – different measures of market reaction.

| Variables                                | CAR1                | CAR2                | CAR3                 | CAR4                 |
|------------------------------------------|---------------------|---------------------|----------------------|----------------------|
| Reach                                    | −0.100<br>(0.205)   | −0.051<br>(0.439)   | −0.121<br>(0.173)    | −0.099<br>(0.222)    |
| Controversial Industry                   | 1.057*<br>(0.071)   | 1.016*<br>(0.070)   | 1.035<br>(0.239)     | 0.544<br>(0.503)     |
| High ESG                                 | −0.067<br>(0.729)   | 0.037<br>(0.838)    | −0.014<br>(0.961)    | −0.015<br>(0.954)    |
| Controversial Industry.<br>High ESG      | −0.880**<br>(0.030) | −0.938**<br>(0.010) | −0.869*<br>(0.059)   | −0.511<br>(0.243)    |
| Reach.Controversial<br>Industry          | −0.347*<br>(0.051)  | −0.343**<br>(0.028) | −0.510**<br>(0.019)  | −0.400**<br>(0.027)  |
| Reach.High ESG                           | 0.007<br>(0.952)    | −0.021<br>(0.827)   | −0.027<br>(0.845)    | −0.043<br>(0.745)    |
| Reach.Controversial<br>Industry.High ESG | 0.584***<br>(0.003) | 0.583***<br>(0.002) | 0.612*<br>(0.073)    | 0.468<br>(0.173)     |
| Breach                                   | −0.157*<br>(0.099)  | −0.042<br>(0.665)   | −0.322***<br>(0.009) | −0.344***<br>(0.006) |
| Severity                                 | 0.003<br>(0.951)    | −0.013<br>(0.795)   | 0.108<br>(0.233)     | 0.125<br>(0.158)     |
| Novelty                                  | 0.127<br>(0.182)    | 0.098<br>(0.285)    | 0.079<br>(0.414)     | 0.097<br>(0.356)     |
| Peak RRI                                 | 0.003<br>(0.484)    | 0.002<br>(0.529)    | 0.005<br>(0.376)     | 0.006<br>(0.223)     |
| Size                                     | 0.008<br>(0.870)    | 0.009<br>(0.853)    | 0.006<br>(0.943)     | 0.009<br>(0.923)     |
| ROA                                      | −1.469*<br>(0.059)  | −0.861<br>(0.131)   | −1.074<br>(0.281)    | −0.947<br>(0.373)    |
| Lev                                      | −0.239<br>(0.272)   | −0.306<br>(0.244)   | −0.454<br>(0.135)    | −0.373<br>(0.238)    |
| PPE                                      | 0.490**<br>(0.029)  | 0.397*<br>(0.094)   | 0.578*<br>(0.084)    | 0.700**<br>(0.036)   |
| Sales Growth                             | −0.374<br>(0.117)   | −0.352<br>(0.180)   | −0.441<br>(0.262)    | −0.251<br>(0.506)    |
| Adj. R-square                            | 0.019               | 0.015               | 0.015                | 0.014                |
| F statistics                             | 1.592               | 1.104               | 1.255                | 1.224                |
| p value (F)                              | 0.003               | 0.272               | 0.09                 | 0.116                |
| Observations                             | 5077                |                     |                      |                      |

Notes: This table presents the results using equation (5) and different measures of CARs. p-values are in parentheses. CAR1 is value weighted CAR (−1,1), CAR2 is CAR(−1,1) using Fama-French-Carhart four factor model (Carhart, 1997), and CAR3 is equally weighted CAR (−2,2), CAR4 is value weighted CAR (−2,2). All equations are estimated without a constant term and include 45 industry dummies variables.

Table 8) is positive and statistically significant. This indicates that stock prices decline more sharply for controversial industry firms with weak ESG performance when negative ESG news receives high (relative to low) media coverage than for their high ESG performance counterparts. We also find that the difference between controversial and noncontroversial industry firms with low ESG performance is significant. Overall, however, the media reach effect is concentrated in firms from controversial industries with low ESG performance.

We then consider the use of high- and low-severity subsamples to assess whether the media reach effect depends on incident severity. Events with severity scores of 2 or 3 are classified as high severity; all others form the low-severity subsample. The pattern of results is similar for both subsamples and, in turn, that pattern is similar to that found for the full sample – except in one regard. A significant negative media reach effect is found for firms from noncontroversial industries with low ESG performance. This effect is smaller in absolute terms than for firms from controversial industries with low ESG performance. The difference is not significant, however. Overall, these results indicate that incident severity conditions the role of media reach, with amplification effects observed in more cases in the presence of high-severity ESG events, consistent with the discussion in Section 3.

We further examine high- and low-share turnover subsamples to assess whether media reach effects are stronger when information asymmetry is lower (the high share turnover subsample). Overall, the

**Table 10**  
Coefficients of Media Reach Using Different CARs.

| CAR1                       |                      |                   |                     |
|----------------------------|----------------------|-------------------|---------------------|
|                            | Low ESG              | High ESG          | Difference          |
| Non-Controversial Industry | −0.100<br>(0.205)    | −0.093<br>(0.280) | 0.007<br>(0.952)    |
| Controversial Industry     | −0.447***<br>(0.006) | 0.144<br>(0.126)  | 0.591***<br>(0.000) |
| Difference                 | −0.347<br>(0.051)    | 0.237<br>(0.070)  |                     |
| CAR2                       |                      |                   |                     |
|                            | Low ESG              | High ESG          | Difference          |
| Non-Controversial Industry | −0.051<br>(0.439)    | −0.072<br>(0.324) | −0.021<br>(0.827)   |
| Controversial Industry     | −0.394***<br>(0.007) | 0.168<br>(0.085)  | 0.562***<br>(0.001) |
| Difference                 | −0.343**<br>(0.028)  | 0.240<br>(0.059)  |                     |
| CAR3                       |                      |                   |                     |
|                            | Low ESG              | High ESG          | Difference          |
| Non-Controversial Industry | −0.121<br>(0.173)    | −0.148<br>(0.172) | −0.027<br>(0.845)   |
| Controversial Industry     | −0.631***<br>(0.002) | −0.046<br>(0.891) | 0.585<br>(0.063)    |
| Difference                 | −0.510**<br>(0.019)  | 0.102<br>(0.770)  |                     |
| CAR4                       |                      |                   |                     |
|                            | Low ESG              | High ESG          | Difference          |
| Non-Controversial Industry | −0.099<br>(0.222)    | −0.142<br>(0.179) | −0.043<br>(0.745)   |
| Controversial Industry     | −0.499***<br>(0.002) | −0.074<br>(0.818) | 0.425<br>(0.179)    |
| Difference                 | −0.400**<br>(0.027)  | 0.068<br>(0.837)  |                     |

Notes: This table presents the coefficients and p-values of F tests for each type of the firms based on the results in Table 9 using equation (5) and different measures of CARs. p-values are in parentheses. The first panel shows the results for full sample using CAR1 in column 2 in Table 9, the second panel shows the result using CAR2 in column 3 in Table 9, the third panel shows the results using CAR3 in column 4 in Table 9, and the fourth panel shows the results using CAR4 in column 5 in Table 9.

results suggest significant media reach effects for all low ESG performance firms with high share turnover. Further, the difference in media reach effects between controversial and noncontroversial industry firms with low ESG performance is significant with the amplification effect lower for the latter. As with the full sample, the media reach effect for controversial-industry firms with low ESG performance differs significantly from and is lower than that for controversial-industry firms with high ESG performance. The corresponding comparison for noncontroversial-industry firms does not produce a significant difference, even though the individual effects are both significant. Low share turnover firms have a similar pattern of results for media reach to the full sample. More media reach effects are found for high (versus low) share turnover firms, consistent with the former firms' share prices reflecting information more efficiently, as suggested in Section 3.

Finally, we examine whether violations of the UN Global Compact constitute an overriding reputational signal that alters the role of ESG performance. The breach subsample reveals pronounced heterogeneity in media reach effects across firms. Although the pattern of media reach effects is similar to that for the full sample, there is also a significant difference between firms from controversial and noncontroversial industries with high ESG performance. For non-breach firm events, there are no significant media reach effects. This pattern suggests that formal norm violations amplify the market's differentiation between firms from

controversial and noncontroversial industries, whatever their levels of ESG performance, as well as between controversial industry firms with high and low ESG performance.

Overall, a generally consistent picture emerges from our results. Consistent with Hypothesis 2b, the results generally suggest that the media reach amplification effect is significantly lower for firms in controversial industries with high ESG performance relative to firms with low ESG performance. The only exception to this result is when we consider the adverse news events that do not involve breaches of the UN Global Compact. Media reach amplification effects vary by sector affiliation and ESG performance and are generally at their strongest for firms in controversial industries with low ESG performance.

Do the results above imply that high ESG performance provides overall insurance to controversial industry firms against negative ESG news, not just against the impact of media reach? To test this, we perform (untabulated) F-tests for different levels of *Reach* for incremental differences in market reactions to negative ESG news events between controversial industry firms with high ESG performance and controversial industry firms with low ESG performance.

To do this, we look at the variables in equation (5) for the full sample that reflect incremental differences in stock market reactions between high and low ESG performance firms in controversial industries. These variables are *Reach.High ESG*, *Controversial Industry.High ESG*, *Reach.Controversial Industry.High ESG* and *High ESG*. For firms with high ESG performance in controversial industries, these variables take values of *Reach*, 1, and *Reach* and 1 respectively. Each of these variables takes the value of 0 for low ESG performance firms in controversial industries. Hence, denoting a coefficient by its variable name, we first evaluate the incremental effect of stock market reactions by testing whether  $(\beta_3 + \beta_4 + \beta_5 + \beta_7)$  is different from zero. This tests whether the incremental effect on stock returns is different from zero for high versus low ESG performance for controversial industry firms for news events for which *Reach* = 1. Similarly, for news events with *Reach* = 2, we test whether  $(2\beta_3 + \beta_4 + 2\beta_5 + \beta_7)$  is different from zero; and for news events for which *Reach* = 3, whether  $(3\beta_3 + \beta_4 + 3\beta_5 + \beta_7)$  is different from zero.

The results suggest that there is no ESG performance insurance effect for news events covered by media with *Reach* equal to 1. Indeed, our estimates suggest that the stock market reaction is worse for firms with high ESG performance in controversial industries than for firms with low ESG performance in controversial industries, all other things being equal. Nonetheless, the size of this effect is not significant other than for the low- and high-turnover subsamples and for news events pertaining to breaches of the UN Global Compact. When *Reach* is equal to 2, the stock market reaction is better for firms with high ESG performance in controversial industries than for firms with low ESG performance in controversial industries, all other things being equal, consistent with an insurance effect being present. In this case, however, the differences are not significant for either the full sample or any of the subsamples other than for news events that do not refer to breaches of the UN Global Compact. When *Reach* is equal to 3, the stock market reaction is again incrementally better for firms with high ESG performance in controversial industries than for firms with low ESG performance in controversial industries, consistent with an insurance effect being present, again, all else being equal. The effects are significant for the full sample and all the subsamples.

Whether high (relative to low) ESG performance provides overall insurance for firms in controversial industries against adverse ESG news depends on the reach of the media outlet covering the story. As identified above, media reach appears to be a moderating variable in determining the overall relationship between high (and low) ESG performance for firms in controversial industries and the stock market reaction to negative ESG news. This moderating effect is of sufficient size to affect whether an insurance effect is seen overall in all circumstances. We can conclude that there is evidence for an insurance effect, at least when the news event is covered by media with high reach. Otherwise, the evidence for a more general insurance effect is weak and, indeed,

**Table 11**

Market reactions to negative ESG media events – using high reach 1.

| Variables                                    | Full sample         | Low Severity        | High Severity       | Low Turnover       | High Turnover        | Breach               | Non-Breach          |
|----------------------------------------------|---------------------|---------------------|---------------------|--------------------|----------------------|----------------------|---------------------|
| High Reach 1                                 | −0.115<br>(0.295)   | −0.038<br>(0.810)   | −0.241<br>(0.120)   | 0.074<br>(0.630)   | −0.254*<br>(0.081)   | −0.134<br>(0.363)    | 0.002<br>(0.987)    |
| Controversial Industry                       | 0.697<br>(0.181)    | 2.977***<br>(0.000) | −0.896<br>(0.218)   | 1.225<br>(0.132)   | 0.296<br>(0.641)     | −0.426<br>(0.569)    | 2.183***<br>(0.008) |
| High ESG                                     | −0.053<br>(0.699)   | 0.050<br>(0.738)    | −0.148<br>(0.467)   | −0.110<br>(0.452)  | 0.053<br>(0.788)     | 0.186<br>(0.352)     | −0.312<br>(0.196)   |
| Controversial Industry.High ESG              | −0.551**<br>(0.049) | −0.683**<br>(0.047) | −0.436*<br>(0.094)  | −0.459*<br>(0.082) | −0.722***<br>(0.001) | −0.683***<br>(0.006) | −0.31<br>(0.173)    |
| High Reach 1.Controversial Industry          | −0.399<br>(0.259)   | −0.488<br>(0.113)   | −0.347<br>(0.438)   | −0.01<br>(0.965)   | −0.901***<br>(0.000) | −0.955***<br>(0.003) | 0.674**<br>(0.043)  |
| High Reach 1.High ESG                        | 0.051<br>(0.781)    | −0.047<br>(0.809)   | 0.214<br>(0.393)    | 0.135<br>(0.465)   | −0.087<br>(0.756)    | −0.154<br>(0.513)    | 0.3<br>(0.274)      |
| High Reach 1.Controversial Industry.High ESG | 0.878***<br>(0.003) | 0.978***<br>(0.009) | 0.742***<br>(0.009) | 0.452<br>(0.145)   | 1.354***<br>(0.001)  | 1.425***<br>(0.000)  | −0.05<br>(0.897)    |
| Breach                                       | −0.104<br>(0.292)   | −0.053<br>(0.612)   | −0.148<br>(0.329)   | −0.006<br>(0.957)  | −0.217<br>(0.151)    |                      |                     |
| Severity                                     | −0.034<br>(0.544)   |                     |                     | 0.033<br>(0.674)   | −0.060<br>(0.544)    | −0.014<br>(0.829)    | 0.008<br>(0.936)    |
| Novelty                                      | 0.112<br>(0.167)    | 0.148*<br>(0.093)   | 0.006<br>(0.966)    | −0.007<br>(0.914)  | 0.246*<br>(0.092)    | 0.173*<br>(0.055)    | 0.021<br>(0.865)    |
| Peak RRI                                     | 0.000<br>(0.936)    | 0.001<br>(0.866)    | 0.001<br>(0.933)    | 0.003<br>(0.298)   | 0.003<br>(0.666)     | 0.001<br>(0.832)     | −0.001<br>(0.933)   |
| Size                                         | 0.021<br>(0.673)    | −0.013<br>(0.809)   | 0.067<br>(0.361)    | −0.117<br>(0.140)  | 0.081<br>(0.237)     | 0.011<br>(0.888)     | 0.036<br>(0.667)    |
| ROA                                          | −1.462**<br>(0.043) | −1.185<br>(0.255)   | −1.528<br>(0.107)   | −0.279<br>(0.814)  | −1.638**<br>(0.044)  | −1.103<br>(0.241)    | −1.628<br>(0.119)   |
| Lev                                          | −0.263<br>(0.252)   | −0.219<br>(0.534)   | −0.236<br>(0.532)   | −0.189<br>(0.679)  | −0.099<br>(0.764)    | −0.433<br>(0.186)    | −0.111<br>(0.734)   |
| PPE                                          | 0.475**<br>(0.044)  | 0.080<br>(0.787)    | 1.028**<br>(0.024)  | 0.057<br>(0.865)   | 0.842**<br>(0.010)   | 0.571*<br>(0.081)    | 0.591<br>(0.332)    |
| Sales Growth                                 | −0.415<br>(0.108)   | −0.01<br>(0.977)    | −1.143**<br>(0.016) | −0.309<br>(0.581)  | −0.423<br>(0.172)    | −0.637*<br>(0.064)   | −0.05<br>(0.907)    |
| Adj. R-square                                | 0.017               | 0.021               | 0.036               | 0.019              | 0.035                | 0.025                | 0.042               |
| F statistics                                 | 1.43                | 1.073               | 1.272               | 0.861              | 1.49                 | 1.363                | 1.486               |
| p value (F)                                  | 0.017               | 0.33                | 0.083               | 0.754              | 0.01                 | 0.037                | 0.011               |
| Observations                                 | 5,077               | 3,006               | 2,071               | 2,542              | 2,535                | 3,083                | 1,994               |

Notes: This table presents the results for the market reaction to negative ESG media coverage using equation (5) and High Reach 1. High Reach 1 is a dummy variable, where High Reach 1 = 1 when Reach = 2 or 3, otherwise equals 0. In this table, column 2 presents the results for equation (5) using full sample. Columns 3 and 4 present the regression results for equation (5) using Low Severity and High Severity sub-samples, columns 5 and 6 present the regression results for equation (5) using the Low Turnover and High Turnover sub-samples, and columns 7 and 8 present the regression results for equation (5) using the Breach and non-Breach sub-samples. p-values are in parentheses. All equations are estimated without a constant term and include 45 industry dummies variables.

under some specific circumstances (breaches of the UN Global Compact), high ESG performance for a firm in a controversial industry is associated with a worse stock market reaction than for one with low ESG performance.

We offer a possible explanation of these results. As mentioned above, for the highest (relative to the lowest) level of media reach, stakeholders, on average, are more likely to interpret the firm's high ESG performance not merely as a risk-mitigating 'insurance' asset but also as a signal of superior managerial foresight, genuine stakeholder commitment, and enhanced resilience. Consequently, they are less likely to penalize a firm with high ESG performance in a controversial industry following adverse news relative to a firm in a controversial industry with low ESG performance.

Compared with prior studies, we provide more refined results regarding any insurance effect of high ESG performance for controversial industry firms. These results suggest that the overall insurance effect of high ESG performance is neither universal nor complete but instead depends on the media reach level when negative ESG incidents are covered by the media.

## 6. Robustness tests

To investigate robustness, we re-estimate Eq. (5) using four alternative cumulative abnormal return (CAR) measures: CAR1 (value-weighted, [−1, 1] window), CAR2 (Fama–French–Carhart four-factor

model), and CAR3 and CAR4 (equally weighted and value-weighted, respectively, over the [−2, 2] window). Tables 9 and 10 report these results and corresponding F-tests.

Untabulated estimations of Eq. (4) yield consistently negative and significant coefficients for Reach across all measures of CAR. The pattern of media reach effects, as with the measure of CAR employed in our main tests on the full sample, suggests that media reach effects are again concentrated in firms from controversial industries with low ESG performance. For the (−1, 1) event window, the differences between the media reach effects for high- and low-ESG-performance firms in controversial industries are significant, but they are not significant for the wider (−2, 2) event window. Overall, the results are not sensitive to the model of normal returns through which abnormal returns are estimated. This is consistent with the view expressed in Kothari and Warner (2007) that the model of normal returns is unlikely to be an issue when an event window as short as (−1, 1) is used. They are sensitive to the use of a larger event window, however, with some result significance being lost. This outcome we attribute to the greater noise introduced into our measure of CAR by employing a longer event window.

In further untabulated tests, we estimate the models using longer event windows of CAR(−5, +5) and CAR(−7, +7). We concentrate on the results for the impact of high versus low ESG performance for controversial industry firms on the effect of media reach. The results are generally weaker, in terms of statistical significance, although of similar magnitude and sign, than those obtained using the shorter event

**Table 12**  
Coefficients of High Reach 1.

| FULL SAMPLE                |                      |                     |                     |
|----------------------------|----------------------|---------------------|---------------------|
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.115<br>(0.295)    | −0.064<br>(0.664)   | 0.051<br>(0.781)    |
| Controversial Industry     | −0.666***<br>(0.009) | 0.263**<br>(0.031)  | 0.929***<br>(0.000) |
| Difference                 | −0.551**<br>(0.049)  | 0.327<br>(0.091)    |                     |
| LOW SEVERITY SUB-SAMPLE    |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.038<br>(0.810)    | −0.085<br>(0.611)   | −0.047<br>(0.809)   |
| Controversial Industry     | −0.721<br>(0.809)    | 0.210<br>(0.144)    | 0.931***<br>(0.005) |
| Difference                 | −0.683**<br>(0.047)  | 0.295<br>(0.179)    |                     |
| HIGH SEVERITY SUB-SAMPLE   |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.241<br>(0.120)    | −0.027<br>(0.860)   | 0.214<br>(0.393)    |
| Controversial Industry     | −0.677***<br>(0.001) | 0.279***<br>(0.007) | 0.956***<br>(0.000) |
| Difference                 | −0.436<br>(0.094)    | 0.306<br>(0.103)    |                     |
| LOW TURNOVER SUB-SAMPLE    |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | 0.074<br>(0.630)     | 0.209<br>(0.137)    | 0.135<br>(0.465)    |
| Controversial Industry     | −0.385<br>(0.074)    | 0.202<br>(0.299)    | 0.587<br>(0.137)    |
| Difference                 | −0.459<br>(0.082)    | −0.007<br>(0.977)   |                     |
| HIGH TURNOVER SUB-SAMPLE   |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.254*<br>(0.081)   | −0.341<br>(0.137)   | −0.087<br>(0.756)   |
| Controversial Industry     | −0.976<br>(0.074)    | 0.291<br>(0.299)    | 1.267**<br>(0.025)  |
| Difference                 | −0.722***<br>(0.001) | 0.632<br>(0.977)    |                     |
| BREACH SUB-SAMPLE          |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | −0.134<br>(0.363)    | −0.288<br>(0.134)   | −0.154<br>(0.513)   |
| Controversial Industry     | −0.817***<br>(0.000) | 0.454**<br>(0.017)  | 1.271***<br>(0.000) |
| Difference                 | −0.683***<br>(0.006) | 0.742***<br>(0.006) |                     |
| NON-BREACH SUB-SAMPLE      |                      |                     |                     |
|                            | Low ESG              | High ESG            | Difference          |
| Non-Controversial Industry | 0.002<br>(0.987)     | 0.302<br>(0.140)    | 0.300<br>(0.274)    |
| Controversial Industry     | −0.308<br>(0.108)    | −0.058<br>(0.853)   | 0.250<br>(0.417)    |
| Difference                 | −0.310<br>(0.173)    | −0.360<br>(0.345)   |                     |

**Note:** This table presents the coefficients and p-values of F tests for each type of firms based on the results in Table 11 using High Reach 1 as the alternative measurement of reach.

windows (other than for the low-severity and low-turnover subsamples). This general pattern might be expected because longer windows are more likely to incorporate confounding firm-specific information unrelated to the focal ESG news event, adding noise to the measure of stock market reaction to the news event. The findings therefore suggest that media reach primarily amplifies investors' immediate reactions to adverse ESG news.

Finally, we also use an alternative measure of reach (*High Reach 1*) to conduct additional tests. *High Reach 1* is a dummy variable, where *High Reach 1* = 1 when *Reach* = 2 or 3, otherwise equals 0. Table 11 presents the results for the market reaction to negative ESG media coverage using Eq. (5) and *High Reach 1*. Table 12 presents the coefficients and p-values of F-tests for each type of firm, based on the results in Table 11.

For media reach effects, Table 12 provides many results that are qualitatively similar to those obtained when *Reach* is used as the measure of media reach. Key differences occur for the low-severity and non-breach of UN Global Compact subsamples, where there are no significant media reach effects for any type of firm. For the full sample and high-severity subsample, there is a further significant difference in results on the influence of media reach on stock returns for firms with high ESG performance in controversial industries. For these firms, the influence of media reach is *positive*. As speculated above, this positive effect of media reach in the presence of adverse ESG news suggests that strong ESG performance in controversial industries can go beyond merely mitigating negative impacts when media attention is high, creating a reputational dividend.<sup>4</sup>

## 7. Conclusions

This study examines how media reach affects the stock market reaction to negative ESG news and, by so doing, revisits the view that high ESG performance universally provides reputational insurance against adverse ESG news events (e.g., Godfrey et al., 2009; Cai et al., 2012; Shiu & Yang, 2017; Afrin et al., 2022; Pandey et al., 2024). Using firm-level adverse ESG events from the RepRisk database, we identify three core findings. First, on average, higher media reach amplifies negative abnormal returns following adverse ESG news, consistent with media agenda-setting theory (McCombs & Shaw, 1972). Second, however, we show that the amplification effect of media reach is not uniform across firms. In particular, the negative impact is mainly concentrated among firms in controversial industries with low ESG performance. This suggests that high ESG performance has an insurance effect for controversial industry firms against the negative impact of coverage by high-reach media following adverse ESG news. Third, our results show that high ESG performance for firms in controversial industries mitigates the adverse effect of negative ESG news only when the news is disseminated by high-reach media outlets. Under lower reach coverage, high ESG performance does not protect firm value in the presence of adverse news. Indeed, for adverse ESG news events related to breaches of the UN Global Compact, high ESG performance is associated with a worse stock return response to the event for firms in controversial industries.

The findings also have clear practical implications. For firms, particularly those operating in controversial industries, ESG engagement can reduce downside risk only if it is sufficiently substantive to withstand high levels of public scrutiny. For stock market participants, the results suggest that the importance of ESG performance in the presence of adverse ESG news should be interpreted in conjunction with media exposure and scrutiny intensity, rather than treated as a stand-alone measure of protection against downside risk from adverse ESG

<sup>4</sup> We reran the analyses using an alternative definition of the High ESG dummy variable, which equals 1 if a firm's ESG performance score is in the top 25% of the distribution and 0 if it is in the bottom 25%. The results remain consistent with those obtained using the original definition of the High ESG dummy. Tables are available from the lead author upon request.



news. For ESG data providers and regulators, the findings highlight the importance of transparency and credible information dissemination.

This study has several limitations. First, although we have removed major contaminating events, other events may still occur on the same date as the ESG news events (such as industry-specific regulatory changes, sector-level crises, or macroeconomic news) that may influence abnormal returns. While these shocks are unlikely to correlate systematically with media reach or ESG performance (De Jong & Naumovska, 2016), they nonetheless introduce residual uncertainty into the interpretation of our estimates. Second, this study focuses exclusively on negative ESG news and does not examine positive ESG events. Positive ESG coverage may offset the impact of controversies, potentially altering how media reach, industry characteristics, adverse ESG news, and firm value interact. In addition, our analysis focuses on short-term market reactions rather than longer-term valuation effects.

Several avenues for future research on the role of media reach in stock market reactions to adverse ESG news remain. For example, delving deeper into the relationship between media reach and the existence of an overall ESG performance insurance effect could be valuable. Future research could also investigate the longer-run consequences of negative ESG news for firm value. Future studies could consider whether firms' voluntary disclosures shape the relationship between adverse ESG news and stock price responses and, conversely, whether voluntary disclosures are differentially shaped by the media reach of adverse ESG news. Other studies could consider supply chain effects of firm-specific adverse news events. Finally, as our sample is limited to U. S. firms, extending the analysis to an international context would help assess the generalizability of these findings.

#### CRediT authorship contribution statement

**Siqi Liu:** Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Wei Jiang:** Supervision, Methodology, Conceptualization. **Andrew W. Stark:** Writing – review & editing, Supervision, Methodology, Conceptualization.

#### Declaration of competing interest

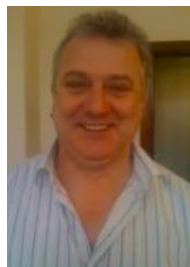
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### References

- Afrin, R., Peng, N., & Bowen, F. (2022). The wealth effect of corporate water actions: How past corporate responsibility and irresponsibility influence stock market reactions. *Journal of Business Ethics*, 180(1), 105–124. <https://doi.org/10.1007/s10551-021-04881-6>
- Alamoudi, M. A., & Hamoudah, M. M. (2023). Do ESG initiatives improve the performance of Saudi-listed banks? The moderating effect of bank size. *International Journal of Accounting, Business and Finance*, 3(1), 30–49. <https://doi.org/10.55429/ijabf.v3i1.137>
- Ali, A., & Ali, F. (2025). External governance monitoring and corporate behavior: Impact of negative media coverage pressure on ESG rating divergence. *Borsa Istanbul Review*, 25(5), 895–907. <https://doi.org/10.1016/j.bir.2025.05.005>
- Bansal, P., & Clelland, I. (2004). Talking trash: Legitimacy, impression management, and unsystematic risk in the context of the natural environment. *Academy of Management Journal*, 47(1), 93–103. <https://doi.org/10.5465/20159562>
- Beccchetti, L., Ciciretti, R., Hasan, I., & Kobeissi, N. (2012). Corporate social responsibility and shareholder's value. *Journal of Business Research*, 65(11), 1628–1635. <https://doi.org/10.1016/j.jbusres.2011.10.022>
- Boone, A., & Uysal, V. B. (2020). Reputational concerns in the market for corporate control. *Journal of Corporate Finance*, 61, Article 101399. <https://doi.org/10.1016/j.jcorpfin.2018.08.010>
- Burke, J. J., Hoiash, R., & Hoiash, U. (2019). Auditor response to negative media coverage of client environmental, social, and governance practices. *Accounting Horizons*, 33(3), 1–23. <https://doi.org/10.2308/acch-52450>
- Cai, Y., Jo, H., & Pan, C. (2012). Doing well while doing bad? CSR in controversial industry sectors. *Journal of Business Ethics*, 108, 467–480. <https://doi.org/10.1007/s10551-011-1103-7>
- Capelle-Blancard, G., & Petit, A. (2019). Every little helps? ESG news and stock market reaction. *Journal of Business Ethics*, 157, 543–565. <https://doi.org/10.1007/s10551-017-3667-3>
- Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of Finance*, 52(1), 57–82. <https://doi.org/10.1111/j.1540-6261.1997.tb03808.x>
- Cui, B., & Docherty, P. (2020). Stock price overreaction to ESG controversies. *SSRN Working Paper*. <https://doi.org/10.2139/ssrn.3559915>
- De Jong, A., & Naumovska, I. (2016). A note on event studies in finance and management research. *Review of Finance*, 20(4), 1659–1672. <https://doi.org/10.1093/rof/rfv037>
- De Vincentiis, P. (2024). ESG news, stock volatility and tactical disclosure. *Research in International Business and Finance*, 68, Article 102187. <https://doi.org/10.1016/j.ribaf.2023.102187>
- Eccles, R. G., Rajgopal, S., & Xie, J. (2022). Does ESG negative screening work? *SSRN working paper*. doi: 10.2139/ssrn.4150524.
- Fafaliou, I., Giaka, M., Konstantios, D., & Polemis, M. (2022). Firms' ESG reputational risk and market longevity: A firm-level analysis for the United States. *Journal of Business Research*, 149, 161–177. <https://doi.org/10.1016/j.jbusres.2022.05.010>
- Fama, E. F., Fisher, L., Jensen, M. C., & Roll, R. (1969). The adjustment of stock prices to new information. *International Economic Review*, 10(1), 1–21. <https://doi.org/10.2307/2525569>
- Fang, L., & Peress, J. (2009). Media coverage and the cross-section of stock returns. *The Journal of Finance*, 64(5), 2023–2052. <https://doi.org/10.1111/j.1540-6261.2009.01493.x>
- Flammer, C. (2013). Corporate social responsibility and shareholder reaction: The environmental awareness of investors. *Academy of Management Journal*, 56(3), 758–781. <https://doi.org/10.5465/amj.2011.0744>
- Godfrey, P. C., Merrill, C. B., & Hansen, J. M. (2009). The relationship between corporate social responsibility and shareholder value: An empirical test of the risk management hypothesis. *Strategic Management Journal*, 30(4), 425–445. <https://doi.org/10.1002/smj.750>
- Grossman, S. J., & Stiglitz, J. E. (1980). On the impossibility of informationally efficient markets. *The American Economic Review*, 70(3), 393–408. <https://www.jstor.org/stable/1805228>
- He, F., Guo, X., & Yue, P. (2024). Media coverage and corporate ESG performance: Evidence from China. *International Review of Financial Analysis*, 91, Article 103003. <https://doi.org/10.1016/j.irfa.2023.103003>
- Kölbel, J. F., Busch, T., & Jancso, L. M. (2017). How media coverage of corporate social irresponsibility increases financial risk. *Strategic Management Journal*, 38(11), 2266–2284. <https://doi.org/10.1002/smj.2647>
- Kothari, S. P., & Warner, J. B. (2007). Econometrics of event studies. In *Handbook of empirical corporate finance* (pp. 3–36). Elsevier. <https://doi.org/10.1016/B978-0-444-53265-7.50015-9>
- Krüger, P. (2015). Corporate goodness and shareholder wealth. *Journal of Financial Economics*, 115(2), 304–329. <https://doi.org/10.1016/j.jfineco.2014.09.008>
- Liaquat, I., Floreani, J., & Naseer, M. M. (2026). ESG performance and bank stability: The role of national culture and formal institutions. *Research in International Business and Finance*, 81, Article 103214. <https://doi.org/10.1016/j.ribaf.2025.103214>
- Lins, K. V., Servaes, H., & Tamayo, A. (2017). Social capital, trust, and firm performance: The value of corporate social responsibility during the financial crisis. *Journal of Finance*, 72(4), 1785–1824. <https://doi.org/10.1111/jofi.12505>
- Lundgren, T., & Olsson, R. (2010). Environmental incidents and firm value—international evidence using a multi-factor event study framework. *Applied Financial Economics*, 20(16), 1293–1307. <https://doi.org/10.1080/09603107.2010.482516>
- McCombs, M. E., & Shaw, D. L. (1972). The agenda-setting function of mass media. *Public Opinion Quarterly*, 36(2), 176–187. <https://doi.org/10.1086/267990>
- Moeller, S. B., Schlingemann, F. P., & Stulz, R. M. (2004). Firm size and the gains from acquisitions. *Journal of Financial Economics*, 73(2), 201–228. <https://doi.org/10.1016/j.jfineco.2003.07.002>
- Nicolas, M. L., Desrozier, A., Caccioli, F., & Aste, T. (2024). ESG reputation risk matters: An event study based on social media data. *Finance Research Letters*, 59, Article 104712. <https://doi.org/10.1016/j.frl.2023.104712>
- Ocasio, W. (1997). Towards an attention-based view of the firm. *Strategic Management Journal*, 18(S1), 187–206. <https://www.jstor.org/stable/3088216>
- Pandey, D. K., Kumari, V., Palma, A., & Goodell, J. W. (2024). Impact of ESG regulation on stock market returns: Investor responses to a reasonable assurance mandate. *Finance Research Letters*, 64, Article 105412. <https://doi.org/10.1016/j.frl.2024.105412>
- Serafeim, G., & Yoon, A. (2022). Which Corporate ESG News does the Market React to? *Financial Analysts Journal*, 78(1), 59–78. <https://doi.org/10.1080/0015198X.2021.1973879>
- Shiu, Y. M., & Yang, S. L. (2017). Does engagement in corporate social responsibility provide strategic insurance-like effects? *Strategic Management Journal*, 38(2), 455–470. <https://doi.org/10.1002/smj.2494>
- Suryani, A. W., Luthfiani, A. D., Restuningdiah, N., Abdul Jalil, A., Sugeng, B., & Pujiningsih, S. (2025). ESG in the spotlight: Does media coverage drive stock market reactions? *Journal of Financial Reporting and Accounting*. <https://doi.org/10.1108/JFRA-10-2024-0748>
- Wong, J. B., & Zhang, Q. (2022). Stock market reactions to adverse ESG disclosure via media channels. *The British Accounting Review*, 54(1), Article 101045. <https://doi.org/10.1016/j.bar.2021.101045>



**Dr. Siqi Liu** is a Lecturer (Assistant Professor) in Accounting and Financial Management at the Open University. Her research interests include market-based accounting research and the economic consequences of corporate social responsibility (CSR) and climate change risk-related information. Her work has been published in the *Journal of Sustainable Finance and Accounting* and the *Review of Quantitative Finance and Accounting*.



**Dr. Andrew Stark** is an *Emeritus* Professor at Alliance Manchester Business School, University of Manchester. He is currently a senior editor of the *Journal of Business Finance and Accounting* and was formerly joint-editor of *The British Accounting Review*. Professor Stark's current research interests include the stock market's demand for, and the consequences of, the disclosure of non-financial information. He has published extensively in leading international peer-reviewed academic journals.



**Dr. Wei Jiang** is currently a Senior Lecturer (Associate Professor) of Accounting at Alliance Manchester Business School, University of Manchester. Her research focuses on the use of accounting information by capital market participants, as well as the effects of corporate governance on firm performance and corporate policies. Her work has appeared in several international peer-reviewed academic journals and in *Forbes* online magazine. Her research interests include market-based accounting research, loss-making firms, dividends, corporate governance, and corporate social responsibility (CSR).